Miniature Passive Pressure-Relief Valve

Low-pressure relief mechanism for long-duration superpressure balloons

Superpressure balloons operate with a sealed, volume-constrained envelope. During flight, solar heating raises the lifting-gas temperature and increases the internal differential pressure. **Day–night thermal cycling** can therefore drive the balloon toward its structural pressure limit. The pressure-relief valve provides passive overpressure protection by venting helium above a defined cracking pressure and resealing as the pressure falls.

1.4–6.5 kPa	−20 °C / −60 °C	±0.9%	27 g / <20 g
ADJUSTABLE CRACKING RANGE	TESTED / TARGET	CRACKING REPEATABILITY, -25 °C	CURRENT MASS / TARGET

ARCHITECTURE

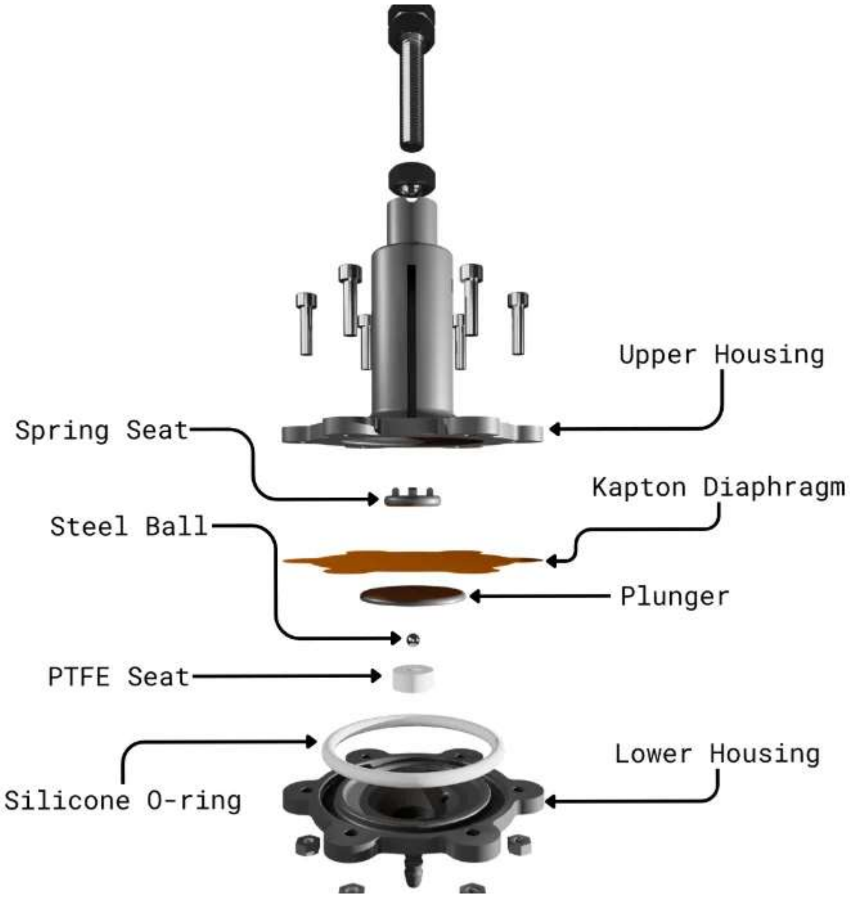


Fig. 1 — Exploded assembly.

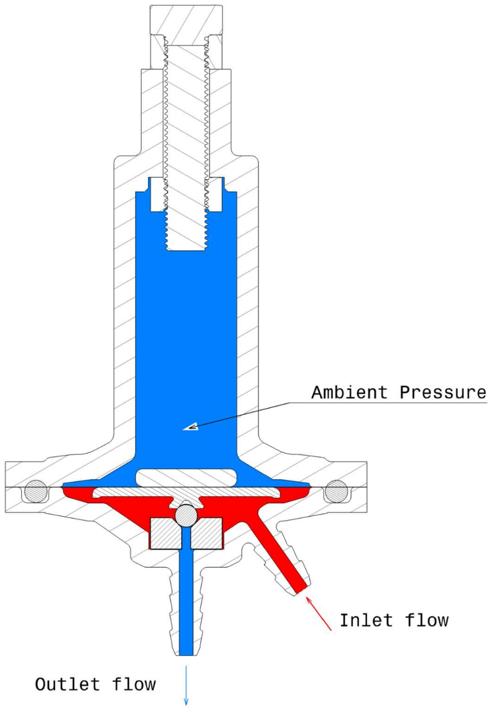


Fig. 2 — Section view: load path from diaphragm through spring and plunger to the steel ball and PTFE seat.

DESIGN CHALLENGES

Low actuation force

Only a few newtons available at 1-6.5kPa-level pressures.

Low-temperature sealing

Elastomer selection becomes difficult toward −60 °C; a steel ball on a machined PTFE seat is used instead.

Leak vs cracking trade-off

More seat load improves sealing but directly raises cracking pressure.

Low mass

Attainable altitude is limited by payload mass, so valve mass must stay under 20 g.

PROTOTYPE

Assembled prototype

Force balance across the diaphragm

At only 1.4–6.5 kPa, the available actuation force is limited to a few newtons. The first direct-acting concept used essentially the same area to generate pressure force and to provide the sealing interface, which led to very low contact stress and made sealing highly dependent on surface finish and lubrication. The diaphragm architecture **separates these functions**: pressure is collected over a large effective area, while the resulting force is transferred to a much smaller ball-seat contact, producing substantially higher local sealing stress.

AREA DECOUPLING

FUNCTION	DIRECT-ACTING CONCEPT	DIAPHRAGM-ACTUATED — CURRENT
Actuation area	SLA-printed resin plunger face	Kapton diaphragm, A_{pred} = 585 mm²
Sealing area	Same face — coincident with actuation	Steel ball on machined PTFE seat
Contact stress	Baseline, 1×	Order-of-magnitude higher — same force over a far smaller contact
Sealing depends on	Surface finish, silicone lubricant	Local contact stress at the ball–seat line

FORCE BALANCE

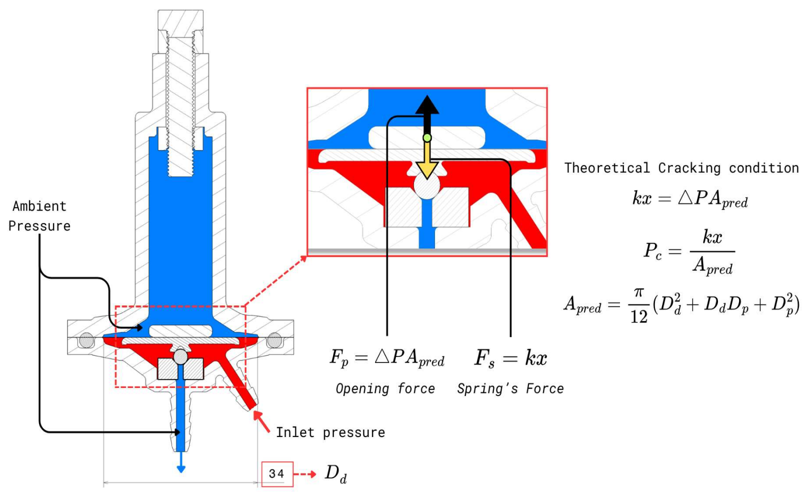


Fig. 3 — Section view. $F_p = \Delta P \cdot A_{pred}$ acts up on the diaphragm; $F_s = kx$ acts down through the plunger onto the ball and seat.

PREDICTED SETPOINT CONTROL FROM THEORETICAL MODEL

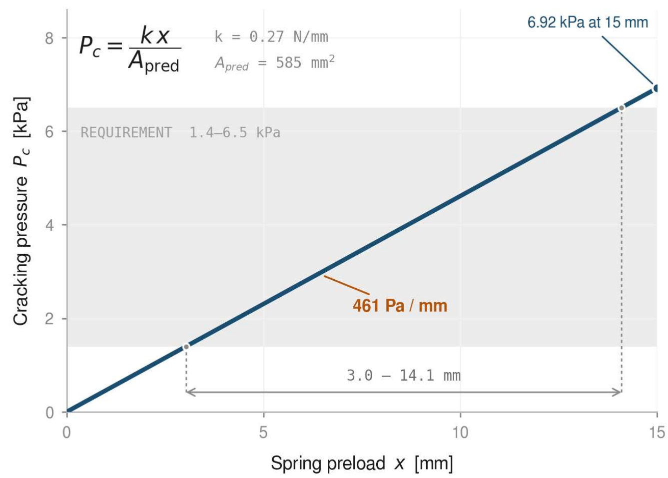


Fig. 4 — Theoretical setpoint control.

461 Pa/mm

SETPOINT SENSITIVITY, $dP_C/dx = k / A_{pred}$

0–15 mm → 0–6.92 kPa

FULL PRELOAD ADJUSTMENT RANGE USABLE KNOWING MAX SPRING COMPRESSION OF 15MM

The area predicted is not the full diaphragm disc: the outer edge is clamped, so only part of the pressure load reaches the centre. Assumptions are based on displacement falling linearly from the bonded Ø20mm plunger to the Ø34 clamp.

CROSS-CHECK

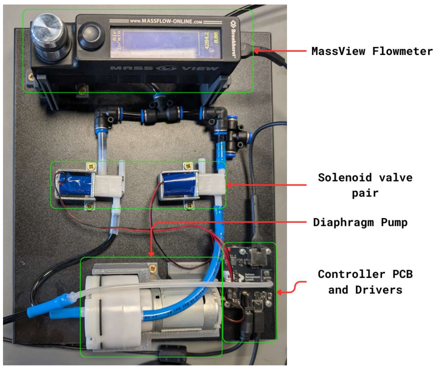
Measured effective area is **599.5 mm²** (page 3), within **2.4%** of this prediction.

CURRENT BUILD

COMPONENT	MATERIAL / SPECIFICATION	FUNCTION
Housing	SLA-printed resin (Formlabs Tough 1.5K)	Structural body
Diaphragm	Kapton, 55 µm incl. acrylic adhesive	Pressure actuator
Spring	Stainless Steel Spring Wire (ISO 6931-1), k = 0.27 N/mm	Setpoint control
Plunger/Spring seat	SLA-printed resin (Formlabs Tough 1.5K)	Transfers diaphragm and spring load
Ball	Hardened steel, G10, ⌀ 3.175 mm	Moving sealing element
Seat	Machined PTFE	Dynamic sealing surface
O-ring	Silicone 45 Sha	Static housing seal

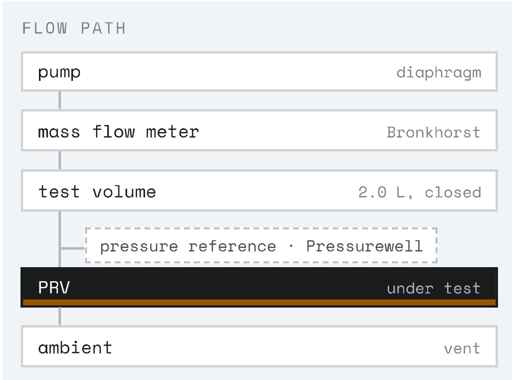
Measuring the valve

TEST RIG



BMS test rig — pump, routing solenoids, flow meter, controller

MicroStep Pressurewell — pressure reference for every run



Automated testing in Python (PyQt5, pandas, SciPy). The rig controls a pump, two routing solenoids and a Bronkhorst mass flow meter, and logs against a MicroStep Pressurewell pressure reference over TCP at approximately 9 Hz.

MEASUREMENT DEFINITION

Cracking pressure

Cracking pressure is the maximum differential pressure reached during pressurization at constant pump command (Constant flow rate).

Leakage

Leakage below the **0.4 slm flow-meter floor** is measured by closed-volume pressure decay, Q = V × |dP/dt|, with the first ~30 s of each step discarded to exclude dead-volume equalisation. Reported with test volume and gas stated, not as a bare Pa/s figure. Test volume is an assumed 2.0 L.

MEASURED CRACKING BEHAVIOUR

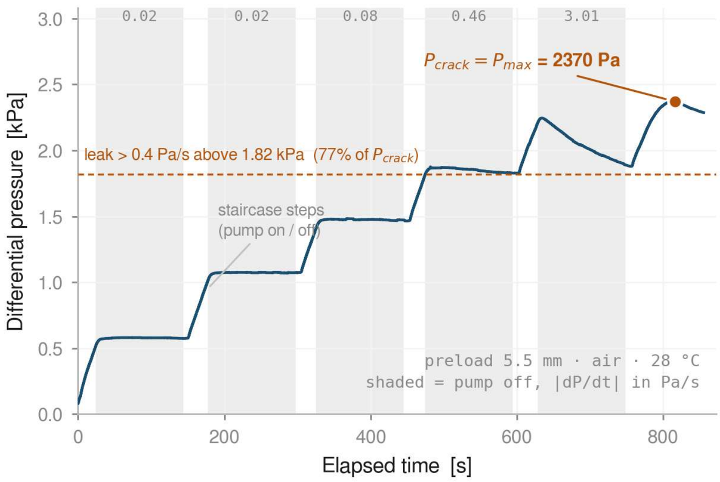


Fig. 5 — Staircase run, 5.5 mm preload, 28 °C, air. Reseat pressure is not measured by this segment.

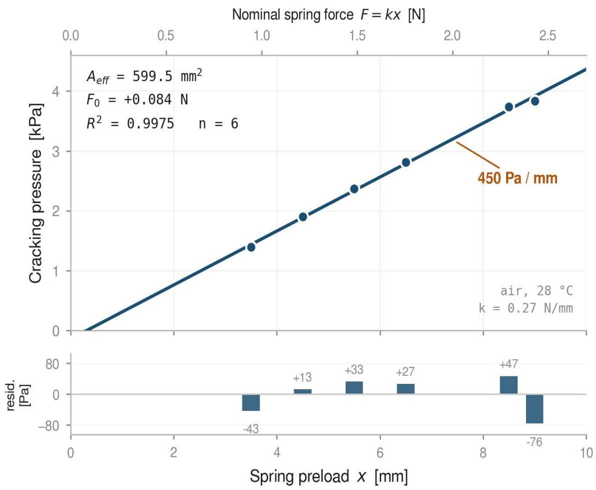


Fig. 6 — Cracking pressure vs preload, Kapton build, air, 28 °C. Fit on n = 6 confirmed cracking events.

AMBIENT CALIBRATION

599.5 mm²	450 Pa/mm	+0.084 N	0.9975
MEASURED EFFECTIVE AREA	SETPOINT SENSITIVITY TO PRELOAD	FITTED OFFSET FORCE F_0	COEFFICIENT OF DETERMINATION, R^2

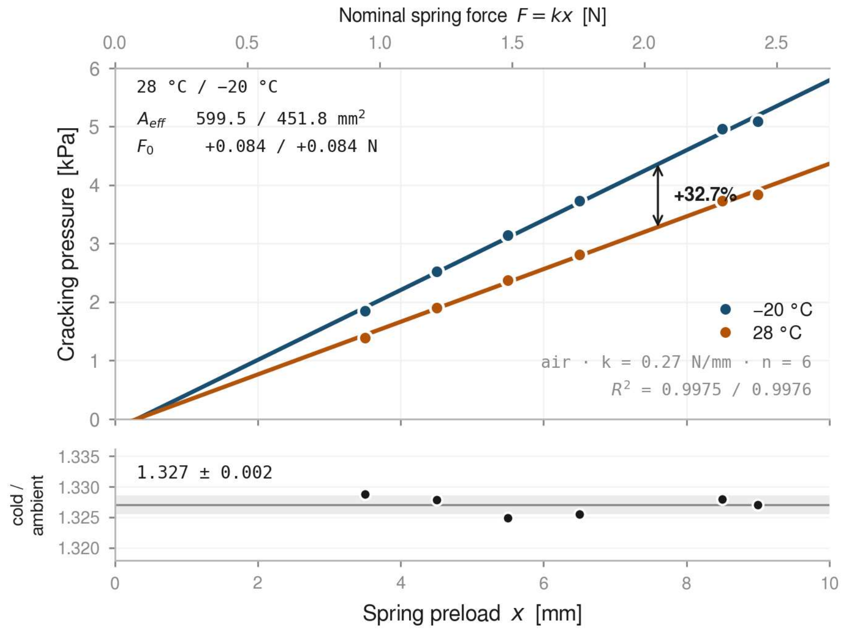
The fitted offset F_0 is a force acting alongside the spring. The earlier Mylar diaphragm gave F_0 = −0.44 N — a parasitic **closing** force of about 734 Pa over the effective area, 52% of the 1.4 kPa low-end setpoint. Kapton reduced it to +0.084 N, but this is not negligible: it is 140 Pa, about 10% of the low-end setpoint. It is also equivalent to 0.31 mm of unaccounted preload, so a residual force and an assembly zero-reference error cannot be separated from this dataset. Direct measurement of spring force at the assembled position would distinguish them. The effective area is measured here, not assumed.

ADDITIONAL RESULTS

Relief capacity	Blowdown	Setpoint repeatability	Closed leakage
Approximately 900 sccm at 4–5 kPa.	[TBD — not yet measured]	Cracking repeatability: 3170 ± 29 Pa (±0.9%) across eight cycles over ten hours at -25 °C.	No leakage was resolved above the ~0.5 sccm at 2.0 L (air) measurement floor until approximately 600-700 Pa below cracking. Background rig leakage has not been independently characterized, so this is an upper bound on combined rig and valve leakage.

Cold behaviour and current limits

DIAPHRAGM — CRACKING PRESSURE VS TEMPERATURE



1.327 ± 0.002

COLD / AMBIENT CRACKING RATIO, CONSTANT ACROSS A 2.6× PRELOAD RANGE

599.5 ⟶ 451.8 mm²

FITTED EFFECTIVE AREA, 28 °C → −20 °C

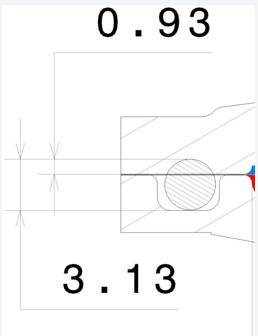
+0.084 N

Fitted offset F_0, identical at both temperatures to within 0.3 mN

1. Cooling to −20 °C raises cracking pressure by approximately 32.7% across the tested preload range.
2. The effect is a constant ratio, not a constant offset. Within the force balance this is a change in gain — effective area — not an additive force.
3. F_0 is unchanged, to within 0.3 mN. An adhesion or stiction force would move the intercept; it does not.
4. Hypothesis, not proven: the acrylic adhesive layer on the Kapton tape stiffens near its glass transition, reducing the freely moving area. A competing explanation is a smooth modulus increase in the diaphragm and adhesive system.

Fig. 7 — Cracking pressure versus preload at 28 °C and −20 °C. The two datasets differ by a constant ratio across the tested preload range, not by a constant offset.

STATIC SEAL — INDEPENDENT COLD FAILURE



Sealing lost at approx. −30 ± 3 °C

Significant leakage develops at the silicone O-ring / housing interface at approximately −30 ± 3 °C. Nominal squeeze at ambient is 29.7%, so the seal is not squeeze-deficient by design. Differential thermal contraction accounts for roughly 6% of nominal squeeze using assumed material properties, and appears too small by itself to explain the gross leakage.

Likely contributors: reduced elastomer compliance at low temperature, clamp-load change, groove and squeeze sensitivity, and compound-specific behaviour. The mechanism has not been isolated experimentally.

Open item: elastomer selection against the −60 °C design minimum.

Fig. 8 — Housing O-ring groove. Values in [mm] Nominal squeeze: 0.93/3.13=29.7%.

STATUS

PARAMETER	REQUIREMENT	MEASURED	METHOD
Cracking range	1-7 kPa	1.4–6.5 kPa	Ramp to cracking, full pump command
Effective area	—	599.5 mm² at 28 °C	Fit slope, n = 6
Relief capacity	>2000 sccm	~900 sccm at 4–5 kPa	Mass flow meter
Closed leakage	Airtight	Upper bound only; below measurement floor (less than ~0.5 sccm)	Closed-volume decay
Operating temperature	−60 °C design minimum	Diaphragm characterised at −20 °C; static seal fails at −30 ± 3 °C	Cold soak
Mass	<20 g	27 g	—

NEXT ITERATION

Diaphragm system

- Intermediate-temperature sweep
- Bonded area and attachment stiffness
- Direct spring-force measurement

Static seal

- Alternative low-temperature elastomer
- Groove and squeeze review
- Clamp-preload investigation
- Possible elimination of the separate O-ring

Qualification

- Helium service
- Thermal cycling
- Endurance cycling
- Final low-temperature qualification

Validated: ambient cracking, cold cracking behaviour at −20 °C | **In progress:** relief capacity increase, static seal below −30 °C, mass reduction | **Planned:** helium, thermal cycling, endurance